

EE114 Homework 3

Homework

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Responses

1. Question 2.2.2

Question: For random variables X and R defined in Example 2.5, find $P_X(x)$ and $P_R(r)$. In addition, find the following probabilities:

- (a) $P[X = 0]$
- (b) $P[X < 3]$
- (c) $P[R > 1]$

Response:

2. Question 2.2.4

Question: The random variable X has PMF

$$P_X(x) = \begin{cases} \frac{c}{x}, & x \in \{2, 4, 8\}, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[X = 4]$?
- (c) What is $P[X < 4]$?
- (d) What is $P[3 \leq X \leq 9]$?

Response:

3. Question 2.2.7

Question: In the ticket agency of Problem 2.2.6, each telephone ticket agent is available to receive a call with probability 0.2. If all agents are busy when someone calls, the caller hears a busy signal. What is the minimum number of agents that you have to hire to meet your goal of serving 95% of the customers who want tickets?

Response:

4. Question 2.3.2

Question: In the ticket agency of Problem 2.2.6, each telephone ticket agent is available to receive a call with probability 0.2. If all agents are busy when someone calls, the caller hears a busy signal. What is the minimum number of agents that you have to hire to meet your goal of serving 95% of the customers who want tickets?

Response:

5. Question 2.3.6

Question: The number of bits B in a fax transmission is a geometric ($p = 2.5 \cdot 10^{-5}$) random variable. What is the probability $P[B > 500,000]$ that a fax has over 500,000 bits?

Response:

6. Question 2.3.7

Question: The number of buses that arrive at a bus stop in T minutes is a Poisson random variable B with expected value $T/5$.

- (a) What is the PMF of B , the number of buses that arrive in T minutes?
- (b) What is the probability that in a two-minute interval, three buses will arrive?
- (c) What is the probability of no buses arriving in a 10-minute interval?
- (d) How much time should you allow so that with probability 0.99 at least one bus arrives?

Response:

7. Question 2.3.8

Question: In a wireless automatic meter reading system, a base station sends out a wake-up signal to nearby electric meters. On hearing the wake-up signal, a meter transmits a message indicating the electric usage. Each message is repeated eight times.

- (a) If a single transmission of a message is successful with probability p , what is the PMF of N , the number of successful message transmissions?
- (b) I is an indicator random variable such that $I = 1$ if at least one message is transmitted successfully; otherwise $I = 0$. Find the PMF of I .

Response:

8. Question 2.3.11

Question: In a packet voice communications system, a source transmits packets containing digitized speech to a receiver. Because transmission errors occasionally occur, an acknowledgment (ACK) or a nonacknowledgment (NAK) is transmitted back to the source to indicate the status of each received packet. When the transmitter gets a NAK, the packet is retransmitted. Voice packets are delay sensitive and a packet can be transmitted a maximum of d times. If a packet transmission is an independent Bernoulli trial with success probability p , what is the PMF of T , the number of times a packet is transmitted?

Response:

9. Question 2.4.2

Question: The random variable X has CDF

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \leq x < 0, \\ 0.7, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

- (a) Draw a graph of the CDF.
- (b) Write $P_X(x)$, the PMF of X . Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞ .

Response:

10. Question 2.4.4

Question: Following Example 2.24, show that a geometric (p) random variable K has CDF

$$F_K(k) = \begin{cases} 0, & k < 1, \\ 1 - (1 - p)^{\lfloor k \rfloor}, & k \geq 1. \end{cases}$$

Response: